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Navigating the power of blockchain strategy: Analysis of technology-organization-environment (TOE) framework and innovation resistance theory using PLS-SEM and ANN insights

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ABSTRACT

Blockchain technology is rapidly replacing traditional technologies. This technology offers numerous benefits, including transparency, safety and security, authenticity, and traceability, which drive business organizations to adopt it. Thus, the factors that generate resistance to the adoption of blockchain technology must be recognized, and an organization's strategy toward adopting blockchain technology must be influenced. In this study, the vital factors that produce resistance to blockchain technology from the perspective of oil companies operating in Iraq were determined. The three-pronged framework of technology, organization, and environment (TOE) was employed to identify the elements that affect resistance to blockchain technology, with subsequent effects on the blockchain strategy. Data were collected using a questionnaire distributed among the managers of the target companies. Among the 381 distributed questionnaires, 313 usable responses were gathered and analyzed via partial least square–structural equation modeling and artificial neural networks to test the proposed hypotheses. The findings show that all the elements of the TOE framework are crucially associated with resistance to blockchain technology and consequently shape the blockchain strategy. Furthermore, resistance to blockchain technology mediates the link between predictors of TOE factors and the blockchain strategy. Therefore, in this work, the present knowledge on resistance to blockchain technology and the blockchain strategy is enhanced by explicating the factors inhibiting blockchain technology adoption in a previously overlooked context. The findings provide practical insight for managers of oil companies to devise effective blockchain strategies for the optimal adoption of blockchain technology.

1. Introduction

Blockchain technology is currently widely used in countless business operations across different sectors; it has far reached business applications and adds value to stakeholders, including suppliers, customers, and partners' digital payments. Blockchain technology can transform the business world when digital documentation, digital identification, transparency, safety and security, efficiency, reduced time and cost, and smart contracts are facilitated. Chittipaka et al. (2023) anticipate that the global investment in blockchain technology will reach 19 billion US dollars by the end of 2024. Accordingly, research on block data from 2021 suggests that 81 out of the 100 top companies with the highest market capitalization opted for blockchain technology. Best (2024) reported that blockchain technology is currently undergoing increasing adoption trends, as evident from a recent report stating that the market size of blockchain technology will grow to 1000 trillion US dollars by

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2032. Similarly, reports from the International Data Corporation (IDC) suggest that global expenditures on blockchain resolutions will reach 19 billion US dollars by the end of 2024. Similarly, the Gartner survey forecasted a global revenue generation of 3.1 trillion US dollars by the year 2030. This massive growth toward the adoption of blockchain technology is not restricted to developed countries; it has also been witnessed in developing countries. According to Dwivedi et al., 2023, the world's renowned companies, such as IKEA, Amazon, Walmart, and Alibaba, cooperated with Microsoft and IBM to incorporate blockchain technology into their business operations. Moreover, nearly 90 % of the companies in three countries, namely, the US, the UK and China, adopted blockchain technology driven by its robust security and copyright protection. These global companies leveraged blockchain technology to streamline their internal processes, enhance supply chain productivity, and develop software. The technology is considered a real game changer offering sustainable and effective solutions to real-world problems. The introduction and proliferation of cryptocurrencies across economies has further amplified the role of blockchain technology. For instance, distributed ledger technology has completely transformed financial sector operations in the financial technology industry. Moreover, the development of digital money has led to remarkable applications of blockchain technology in recent years. Several countries are introducing their central bank digital currencies to facilitate international transactions. Cole et al. (2019) reported that, beyond the financial sector, blockchain technology has widespread implications for supply chains and retail businesses.

Angelis and Da Silva (2019) defined blockchain as a distributed ledger of the transactions between the participating partners in a secured tokenized form. Zhou et al. (2020) defined blockchain technology as a database of all digital transactions, records, and events that are formulated, executed, and shared among an assortment of multiple parties. Blockchain contains features such as enhanced safety and security with the provision of decentralized shared services. The safety and security features prevent unauthorized access to the data and prohibit data alteration. Hackius and Petersen (2017) and Tapscott and Tapscott (2017) reported that the information stored in a database can be easily traced and sorted, enabling the transparent conduct of audits for well-informed business decisions. Chang and Chen (2020) argued that blockchain technology helps businesses ensure rapid transactions.

Various factors contribute to the widespread growth of blockchain technology, including supportive government policies, recent market trends, rising demand for customized products, the combination of blockchain with other technologies such as the Internet of Things, artificial intelligence, and resource optimization. Blockchain technology, despite its several benefits to businesses in many ways, is prone to certain potential challenges and drawbacks. The corollary limitations and hindrances must be considered prior to gaining the benefits of using blockchain technology. The implementation and usage of blockchain technology raises multiple concerns at both the functional and the psychological levels. For example, as reported by Dwivedi et al. (2023) and Tan and Salo (2023), data security and data protection are crucial issues that challenge the functional and psychological operations of blockchain technology; they can deter the adoption of blockchain technology, which is essential to the development of a blockchain strategy. Similarly, a lack of appropriate knowledge and skills, interoperability, safety and security, high energy consumption, and low scalability are other major issues that prevent the adoption of blockchain technology. In addition to these issues, Zhou et al. (2020) identified other key indicators that contribute to the failure or success of blockchain technology adoption. The indicators include the cost of developing and maintaining the technology, high energy consumption, and the enormous financial investment needed for the development of private blockchain. In addition, reliability concerns, insufficient expertise, and the absence of digitization in the supply chain were viewed as major challenges to the adoption of blockchain technology. Tapscott and Tapscott (2017) recommended the involvement of senior management and experts to overcome potential challenges in the adoption of technology.

The abovementioned discussion on the benefits and allied issues of blockchain technology indicates that although blockchain offers numerous advantages, it is limited by drawbacks. Blockchain technology can face serious challenges that hinder its adoption. For example, the introduction of a new technology is always complex, and it faces resistance (Choi et al., 2020). Some companies quickly accept new technology, whereas others resist it for a long time to decide on the adoption of new technology. Scalability, governance, and the absence of policy are the major factors that hinder blockchain technology adoption in the financial sector. Similarly, Radanović and Likić (2018) posited that blockchain is relatively less developed and that adopting a novel technology such as blockchain is always risky. In some scenarios, organizations are unprepared to embrace new technology. The adoption of new technology always requires changes in organizational processes, as the existing organizational culture is seldom supportive of new technology. Until new technology is widely adopted, blockchain technology may still be contested. In addition, Goel and Neduncheliyan (2023) argued that recent blockchain technology has significant concerns, such as high-time consumption and low confidentiality, due to harmful attacks and unstructured records. Subsequently, organizations need to develop a blockchain strategy for the fruitful adoption of blockchain technology to meet security threats and ensure data integrity.

The increasing global adoption of blockchain technology has sparked researchers' interest in investigating the field further. In particular, the wider business applications of blockchain technology are offering researchers many opportunities to investigate the adoption of blockchain technology from different perspectives. Further research is needed to explore the aspects that influence the adoption of blockchain technology across different business sectors to embrace technological innovations. In addition, the rapid adoption of blockchain technology requires continuous research efforts to unveil the potential benefits and associated challenges with blockchain technology (Krichen et al., 2022; Stratopoulos et al., 2022; Zhou et al., 2020). For this purpose, the present study attempts to elucidate the impact of technology-related factors (relative advantage, compatibility, trust, security) and organization-related factors (lack of technological knowledge, environment, rivalry pressure, business partner pressure, regulatory support) on resistance to blockchain technology and consequently on the blockchain strategy. In this study, oil companies operating in Iraq are explored, as they belong to a less explored context despite their prominent issues pertaining to blockchain technology. The research findings are expected to enhance the theoretical understanding and generate potential practical implications for blockchain technology in a relatively novel context.

2. Literature review

2.1. Blockchain strategy

Blockchain, one of the most popular technologies during its first few years of application (Chittipaka et al., 2023), was first established by Satoshi Nakamoto in 2008 and used to inhibit double spending, which Chittipaka et al. (2023) and Dehghani et al. (2022) described as “a unique problem in cryptocurrencies.” Nakamoto (2008) described Bitcoin as a peer-to-peer automatic cash system that uses digital signatures to timestamp each transaction structured on a distributed ledger called blockchain technology. Despite the widespread usage of Bitcoin as a blockchain, Tao et al. (2022) argued the importance of distinguishing between cryptocurrency and blockchain. Iansiti and Lakhani (2017) characterize blockchain as a type of technology that increases the efficiency of recording, tracking and executing transactions in a streamlined, highly efficient, transparent and trustworthy manner. Dwivedi et al. (2023) described the cryptographic properties of blockchain as constituting the core of blockchain technology, which uses hashes and cryptography to sign transactions. Blockchains are identified by

cryptographic hashes connected to hashes of the previous blocks to form a chain. Kamble et al. (2021) argued that the information stored on the blockchain is immutable and possesses high transaction security. Wang et al. (2022) suggested that smart contracts can be implemented safely and reliably via blockchain.

Many researchers (e.g., Gong et al., 2023; Kashem et al., 2023; Kouhizadeh et al., 2021; Thompson and Rust, 2023; Tseng et al., 2023) have become increasingly interested in the application and adoption of blockchain. Babich and Hilary (2019) explored weaknesses and challenges and suggested further research to apply blockchain in supply chains. Azzi et al. (2019) investigated the impact of blockchain on supply chains and concluded that technology has an impact on the authenticity, transparency, and security of supply chains. Shen et al. (2022) explored the impact of blockchain on combating and reducing counterfeit competitors in supply chains. Garaus and Treiblmaier (2021) explored the possibility of blockchain to help retailers increase sustainability and enhance the retail channel ecosystem. Dwivedi et al. (2023) claimed that blockchain is the most reliable technology, supporting the findings of Wang et al. (2022), who reported the blockchain's ability to store data safely, provide transparent, flexible, protected business processes, and decrease transaction costs. Nonetheless, some users have expressed concern that their privacy will be violated through audits. In this context, Dwivedi et al. (2023) and Wong et al. (2023) identified security challenges in blockchain technology, arguing that its effectiveness remains underevaluated. Yli-Huumo et al. (2016) noted that blockchain's immaturity and dependence on new tools for its implementation generate resistance to its adoption. In another study, Wong et al. (2023) questioned the psychological readiness of managers to adopt immature technologies on the basis of individual-to-individual differences. Hence, research on barriers related to the application of blockchain in organizations is crucial to provide better evidence and insights into these obstacles.

2.2. Resistance to blockchain technology

According to Jasperson et al. (2005) and Walsh et al. (2021), resistance serves as an inhibiting force that impedes the adoption and accommodation of new behaviors in functional systems. In view of the abovementioned discussion, Adel and Younis (2023) and Chan et al. (2023) argued that understanding the nature of resistance to the blockchain strategy is important for researchers in information systems who seek to explain the lack of innovations, especially when blockchain technology is emerging as an essential feature of organizations. Thus, understanding the factors that influence blockchain resistance provides a clear roadmap for stakeholders to deliver more effective blockchain systems in the future, as emphasized by Munim et al. (2022) and Shishan et al. (2022). Furthermore, according to Conoscenti et al. (2017) and Muzammal et al. (2019), resistance to blockchain generally refers to the inability of blockchain technology to withstand interference and maintain the security of the blockchain architecture by ensuring the immutability of monitoring information. However, Dwivedi et al. (2023) and Friedman and Ormiston (2022) reported that resistance is rather a broader technical notion than simply referring to the unwillingness or refusal of organizations and individuals to use the technology. Status quo bias theory suggests that people resist new technologies because they are biased in favor of current and existing technologies. Theoretically, users can evaluate the costs and benefits of switching to new technologies compared with the status quo (Huang et al., 2021; Samuelson and Zeckhauser, 1988; Kim and Kankanhalli, 2009). Technological change is often viewed as an undesirable resource, whereas established practices and uncertainty drive the status quo (Shahzad et al., 2024). Thus, as viewed by Friedman and Ormiston (2022) and Komulainen and Nätti (2023), the barriers (e.g., psychological and functional barriers) that cause rejection of innovation and technology can adapt to the organizational context.

Antioco and Kleijnen (2010) and Joachim et al. (2018) argued that psychological barriers stem from managers' inherent beliefs and perceptions about barriers that directly or indirectly influence resistance to the adoption of blockchain. In this context, according to Komulainen and Nätti (2023), risk is one of the utmost examined barriers in blockchain adoption and acceptance models. In fact, all new technological innovations pose risks and have potentially unpredictable side effects, as reported by Kautish et al. (2023). Executives' decisions are generally shaped by the perception of risk. For example, as reported by Komulainen and Nätti (2023), when managers perceive that adopting an innovation threatens the smooth operation of existing procedures, especially because it conflicts with the current usage patterns of the organization and the system, it may negatively affect costs and productivity. In addition, implementing a new technology requires investing both time and financial resources (procuring the necessary infrastructure for new technology implementation). Mahmood et al. (2021) and Talke and Heidenreich (2014) found that an organization's management may face challenges owing to financial constraints in implementing new technology. If the technology is in the budding stage, then the organization's management may assess the period required to achieve benefits; long-term procurement needs enormous investments and complex implementation processes, and the benefits may take longer to appear. Furthermore, Komulainen and Nätti (2023) claimed that evaluating advantages in financial terms is a complex process, and it may generate uncertainty and financial risk.

Joachim et al. (2018) and Ram and Sheth (1989) reported that psychological barriers emerge when an organization's management perceives significant changes associated with adopting new technology, such as complexity in integration with the existing system and processes that may deter successful adoption. Komulainen and Nätti (2023) argued that deviation from established traditions and operational processes constrains an organization's management, particularly when it fails to perceive value in adopting the innovation. Decision makers resist switching to blockchain from existing platforms because of uncertainty about the benefits of shifting and practitioners' reservations about profits from investment stipulated or when the cost of operating a double scheme is unjustifiable. These factors can negatively influence the perceived value of the adoption of innovation and change, as reported by Antioco and Kleijnen (2010) and Parasuraman and Grewal (2000). In addition, according to Iyanna et al. (2022) and Joachim et al. (2018), when users find blockchain difficult or incomprehensible, it may cause resistance. Dehghani et al. (2022) and Mahmood et al. (2021) claimed that the absence of skills, inexperience, and inability to work with a new technology prevent adoption in its early stages. Fawcett et al. (2008) and Sayogo et al. (2015) further found that employees lack the competence and skills to perform organizational and environmental roles and tasks, thus causing frustration and hesitation in the use of advanced technologies. Finally, management may also resist adoption when they are concerned about how blockchain can negatively impact the operations for which they are responsible. Therefore, as reported by Friedman and Ormiston (2022), management may defer its adoption, especially because blockchain technology requires business processes and routine changes. Consequently, new technologies require a relatively longer period to gain acceptance.

2.3. Innovation resistance theory

Innovation resistance theory (IRT) is one of the most useful approaches for understanding customers' resistance to innovation. The theory was first postulated by Ram (1987), and later, Ram and Sheth (1989) expanded its scope, and Hew et al. (2019) argued that resistant behavior results from customers' intrinsic desire to maintain the status quo. Innovations are considered disruptive and challenging to belief systems entrenched in the status quo. Consumers are hesitant to buy innovative products because such products can alter their lifestyles, resulting in resistance behavior (Ram and Sheth, 1989). In this context, the measure of resistance to innovation is critical for determining the

success of innovation. Kaur et al. (2021) described two contradictory aspects of customers' resistance to innovation, i.e., active resistance and passive resistance. Active resistance is caused by the characteristics or features of the innovation, and it is referred to as a functional barrier to innovation. Functional barriers are related to the value, usage, and risk of innovation, resulting in resistance to innovation. However, according to Ram and Sheth (1989), passive resistance is caused by conflicting situations resulting from a change in the belief system. The value barriers measure consumers' resistance to innovation in terms of facial value, which persuades them to believe the lack of difference in the value of the existing product and the innovation. Usage barriers indicate that resistance results from the incompatibility of innovation with existing experiences, habits, and lifestyles. Furthermore, risk barriers suggest that resistance occurs because of the perceived physical, social, and economic risk of using the innovation. Chen et al. (2022), who investigated the factors of mobile ticketing applications, identified three major factors that cause customer resistance behavior, namely, innovation characteristics (compatibility, complexity, relative advantage, and perceived risk), consumers' psychological characteristics (perception, motivation, and experience), and propagation mechanisms (credibility, transparency, and information sources).

IRT provides a comprehensive understanding of resistant behavior toward innovation, as it postulates the factors (value, use, risk, image, or tradition) that pose resistance. Kaur et al. (2021) and Sadiq et al. (2021) reported that the IRT has become significant in unlocking customers' behavior as innovations continue to enter the market. Many studies (e.g., Chen et al., 2022; Sadiq et al., 2021) have shown that IRT remains a priority among scholars in explaining innovation resistance behavior. In this work, resistance to blockchain from the perspective of innovation resistance theory is investigated. In previous studies, the perspective of IRT has been used in terms of functional and psychological barriers to encompass customers' resistance to innovation. However, their main concern remained understanding resistance from the end-user perspective. According to Dwivedi et al. (2023), resistance from the perspective of business customers has been a largely overlooked area, especially with respect to blockchain technology. The current study conceptualized relative advantage, compatibility, and firm size as functional barriers. In this study, concepts such as trust, security, lack of knowledge, rivalry pressure, business partner pressure, and regulatory support are analyzed as psychological barriers to blockchain technology. Thus, in the current research, IRT is believed to be far superior in explaining resistance to blockchain.

2.4. Technology, organization, and environment theory

The utilization of digital equipment such as blockchains is still in its burgeoning stages (Ahmad et al., 2022; Munim et al., 2022). Blockchain is currently extensively explored in various contexts; one promising area of investigation is its application in the global supply chain (Xu et al., 2023). Lu et al. (2021) examined blockchain technology in the senior care sector. Kabra (2023) investigated the implementation of blockchain in the Indian healthcare industry. Ahmad et al. (2022) examined the uses and difficulties of blockchain technology in the oil and gas industry by using a systematic literature review design. The extant literature (e.g., Alnoor et al., 2024a; Komulainen and Nätti, 2023; Walsh et al., 2021) on technology implementation, which uses diverse and multiple theoretical models, has focused more on information technology acceptance in organizations. These theoretical models investigate the factors that influence individuals, such as customers and employees, in their technology adoption decisions in terms of different theories, such as technology acceptance theory (Davis, 1989), the theory of reasoned action (Ajzen, 1985), and the diffusion of innovation theory (Rogers et al., 2014). Conversely, theories classified at the organization level analyze the factors that influence adoption decisions at the organizational level (Tornatzky and Fleischer, 1990). One such perspective is the technology, organization, and environment (TOE) theory proposed by Tornatzky and Fleischer (1990). The underlying notion is that the factors within the three contexts that influence organizational adoption decisions determine whether an organization will adopt an innovation. As blockchain technology is a strategic effort at the organizational level (Kabra, 2023), the TOE framework is ideally suited for investigating the determinants of blockchain technology.

The TOE framework of Tornatzky and Fleischer (1990) successfully identified innovation adoption drivers (Tseng et al., 2023). The different TOE elements collectively impact organizational decisions. These elements have more power and accuracy in assessing the acceptance of innovation in the organizational context (Katebi et al., 2022). Many previous researchers (e.g., Gupta et al., 2022; Kamble et al., 2021; Ngah et al., 2022; Tseng et al., 2023), who used the TOE framework to investigate various technical advancements and technologies, have suggested that the framework is useful in examining organizational-level adoption, and substantial empirical evidence has been obtained. Ngah et al. (2022) argue that the development of this framework has filled the void in the established theories of adoption and provided a broader context for the adoption of innovation in an organizational context. In contrast to other adoption models and theories, the TOE framework is a flexible theory that allows the expansion and inclusion of many factors and categories of influences, regardless of the elements to be considered for investigation.

The framework of TOE allows researchers to add other dimensions to rationalize the adoption of innovations. Using this framework, many researchers (e.g., Tariq et al., 2021; Zhu et al., 2005) arbitrarily select variables aligned with the context of the study, even when they are different from those already studied. Despite the varying contexts the three factors represent, the literature review shows that the TOE framework offers a sufficient explanation for managerial decisions to adopt new technologies. According to (Gupta et al. (2022) and Katebi et al. (2022), the framework has received enough theoretical and practical support to be extensively used in the adoption of new technologies across various fields. Taking the lead from former studies, TOE is used in this work as a theoretical lens to realize the research objectives. In addition, the current study's comprehensive framework better explains the issue to be studied. In the current study, the three elements of the framework should be understood in certain specific contexts: the technological context is defined as relative advantage, compatibility, trust, and security. As argued by Chittipaka et al. (2023), Dehghani et al. (2022), and Dwivedi et al. (2023), the organizational context describes a lack of technological knowledge and firm size, and the environmental context is defined by rivalry pressure, business partners, and regulatory support.

2.5. Hypothesis development

2.5.1. Technological elements and resistance to blockchain

As reviewed by Tseng et al. (2023), Yuen et al. (2021), and Rogers (2003), the technological dimension includes four elements: relative advantage, security, trust, and compatibility. Comparative advantage refers to the benefits (enhanced performance and productivity) achieved after the implementation of blockchain technology. Technological innovation provides more intangible benefits to organizations (Friedman and Ormiston, 2022; Kant, 2021). This relative advantage is referred to as a critical factor related to the adoption of information technology (Chittipaka et al., 2023; Ghode et al., 2020), the adoption of cloud computing (Alharbi et al., 2016), and the supply chain (Wong et al., 2020). In this context, relative advantage is a critical factor influencing an organization's adoption of innovative technologies, including blockchain technology. According to Yuen et al. (2021), compared with other traditional technologies, the implementation of blockchain in companies provides the ability to trace products and provide accurate and transparent information about products. Thus, the relative advantage provides a high perception of the ease of using modern technologies, and enormous benefits are achieved via

blockchain in corporate operations. Kamble et al. (2021) reported that companies in different sectors are adopting blockchain technology because of the superior advantages that such technologies bring to companies compared with traditional or current systems.

Alnoor et al. (2024b) and Kamble et al. (2021) found a negative relationship between compatibility and resistance to innovation. Compatibility refers to the ability of a new technology to adapt and integrate with previous organizational practices, processes, procedures, information technology infrastructure, and networks associated with the technology to be used by businesses (Choi et al., 2020). The ease of employing modern technologies is linked to the degree of compatibility and integration of the adopted technology with the company's procedures (Oliveira et al., 2016). Dutta et al. (2020) and Khanna and Haldar (2022) reported that compatibility and resistance to new technologies are negatively correlated. For example, the innovation or technology to be used in business models is examined by practitioners and decision makers to determine the degree of compatibility of such technologies and systems with traditional systems. Shahzad et al. (2024) claim that integrating new technologies with traditional systems is more suitable than replacing traditional technologies to adapt the organization's practices and systems to modern methods. On this basis, according to Han and Chen (2021) and Kamble et al. (2021), the compatibility of blockchain with the information technology infrastructure of organizations increases the likelihood of successful implementation and employment of innovative technologies in companies. The incompatibility of blockchain with organizations' policies and processes is a major barrier and deterrent to the use of advanced technologies such as blockchain or the metaverse.

Garaus and Treiblmaier (2021) describe trust as another factor that is primarily considered when evaluating technology acceptance; conversely, Francisco and Swanson (2018) view trust as the tradeoff between cost and benefit concerning the risks associated with adopting a technology. Previous studies suggest that blockchain-based tracking systems provide high traceability and transparency (Ahmad et al., 2022; Chew et al., 2024) and high levels of trust (Montecchi et al., 2019); these researchers theoretically and empirically determined the positive relationship between trust and blockchain technology. Chittipaka et al. (2023) and Lu et al. (2021) confirmed that customers experience greater trust in companies that implement a blockchain-based tracking system than in those that implement a traditional tracking system. Trust is proposed as a concept in which organizations are willing to adopt blockchain technology because of its conviction in the services that the technology can provide. In general, as explained by Shahzad et al. (2024), a lack of security and confidentiality in organizations increases the risk of compromising the privacy of corporate information and exposing data systems to privacy breaches. Therefore, security and privacy concerns increase the resistance of organizations to implementing technologies (Kouhizadeh et al., 2021). Insecure users are skeptical about sharing data or information, especially since they view technology as risky (Tseng et al., 2023). Furthermore, Shahzad et al. (2024) stated that blockchain technology stores intelligent, contract information, suggesting that blockchain communications are secure. Therefore, organizations can benefit from blockchain technology by increasing customer trust and reducing security concerns, as explained by Kouhizadeh et al. (2021). Behnke and Janssen (2020) found that the security and privacy of personal and organizational information increase the trustworthiness of blockchain technology and eschew the risks of online attacks, fraud, or deviant behavior by increasing the transparency of regulatory processes, especially since blockchain technology encrypts data. Feeling protected when transacting via blockchain technology essentially reduces concerns about its use. Hence, technological elements are assumed to positively impact the adoption of blockchain technology.

Hypothesis 1. (H1)*:* The technological elements (i.e., relative advantage, compatibility, trust, and security) of blockchain technology are related to its resistance.

2.5.2. Organizational elements and resistance to blockchain

Alalwan (2020) found that the absence of technological knowledge is the perception of a lack of knowledge and skills necessary and sufficient to use new technologies. If individuals in an organization possess the necessary expertise and skills, then they are more likely to adopt innovations in technology; conversely, if they lack knowledge and skills, then they are more likely to resist the adoption of new modes of digitization (Dehghani et al., 2022). Dehghani et al. (2022) and Kuan and Chau (2001) showed that companies that use electronic data interchange technology have superior operational efficiency and technical knowledge than do companies that have not adopted such technology. The increasing lack of knowledge about new technologies such as the metaverse or blockchain is a catalyst for lower intentions to adopt new technologies. Kaur et al. (2021) indicated that a lack of required knowledge and skills reduces the likelihood of innovation adoption. Anwar et al. (2021) claimed that technical knowledge and skills increase the dedication of companies and users to adopting modern technological innovations. Chen et al. (2022) asserted that the level of knowledge about innovation and its adoption are positively related. According to Dehghani et al. (2022) and Mioduser and Betzer (2008), most companies have some technical knowledge of one or more recent innovations. However, as explained by Dehghani et al. (2022), organizations with less availability of the required technical knowledge find it difficult to develop and implement innovation, resulting in a lower intention to adopt innovation. Investigating the level of technological knowledge and blockchain is urgently needed for research exploration theoretically and for industries practically. Inevitably, the growing lack of knowledge toward new technologies affects the intention to use them and thus increases resistance to the use of emerging new technologies.

Chittipaka et al. (2023) and Tseng et al. (2023) demonstrated that firm size (number of employees and asset size) is the second critical organizational factor in this research and is referred to as the primary resource of an organization. The number of employees is a decisive factor in determining the size of a company. For example, companies with more than 500 employees are considered large companies, whereas companies with fewer than 500 employees are considered small companies. As further explained by Knott and Vieregger (2020), companies seek to maintain a small size, which provides an advantage to large companies related to decentralization and to small companies as an edge of centralization. Furthermore, the size of companies is a barrier to the use of advanced technologies. Large firms have sufficient human and capital resources and conduct research and development more effectively; they also have greater technological diversity, which enables them to address unexpected problems and complex issues (Shahzad et al., 2024). Furthermore, as explained by Ahmadi et al. (2017), medium- and small-sized firms are willing to adopt new technologies to meet customers' needs and requirements, but their limited capabilities and resources limit their ability to adapt to innovations. In this context, previous studies have shown that firm size positively affects a firm's ability to adopt and use technology (Shahzad et al., 2024; Tseng et al., 2023). According to Clohessy and Acton (2019), large organizations are more likely to adopt and utilize blockchain technology than small organizations are, which may be attributed to the effective research and development efforts that large organizations undertake to adopt innovations. In addition, larger firms are eager to adopt innovations because they are better equipped to adapt to risks than smaller firms are (Chittipaka et al., 2023). Therefore, firm size is hypothesized to be essential in firms' intentions to adopt new innovations and gain competitive advantages in the market.

Hypothesis 2. (H2)*:* The organizational elements (i.e., lack of technological knowledge and firm size) of blockchain technology are related to its resistance.

2.5.3. Environmental elements and resistance to blockchain

The effects of three key environmental elements—rivalry pressure, regulatory support, and business partners—are investigated in this research. Chittipaka et al. (2023) and Gökalp et al. (2019) denote rivalry pressure as the competitive forces that organizations face from rival organizations operating within the same industry. Thompson and Rust (2023) found that firms need to match or excel in their rival's efforts to gain market share, which produces inherent pressures on firms as they develop their innovative capabilities in response to competitors' actions. In this context, as explained by Han and Chen (2021), given that competitors implement new technological practices, they entice other organizations in the same industry to adopt similar practices. Dutta et al. (2020), Kamble et al. (2021), and Kouhizadeh et al. (2021) reported that rivalry pressure is a crucial factor influencing the adoption of technology. Therefore, rivalry pressure is assumed to be one of the key drivers of technological innovation for organizations, especially since organizations always seek to maintain intense competition that guarantees them a distinguished position in the market.

Alharbi et al. (2016) explained that a distinct manifestation of rivalry pressure is partner pressure, which refers to the pressure an organization faces from its business partners. Francisco and Swanson (2018) characterized partners' cooperation as essential and crucial to sustaining business relationships. The use of new technologies requires strong cooperation and cohesion among business partners. Many practitioners confirm that concurrent engineering ensures that different parties work in a unified spirit. According to Malik et al. (2021), the absence of financial and regulatory parties and the lack of technical support exacerbate the problems of adopting and using new technologies. Therefore, complementing the work of Baral and Verma (2021), pressure from business partners is an important contributor to and motivator of technology acceptance. The use of blockchain technology increases the probability of companies surviving and achieving a superior competitive advantage that ensures superiority and innovation over competing companies. However, regulatory support is considered a catalyst for resistance to the use of innovative technologies. Chittipaka et al. (2023) found that regulatory support refers to the policies, rules, regulations and laws enacted by governments to regulate and monitor various industries in terms of the use of new technologies. Baral and Verma (2021) demonstrated that government intervention and legislation are essential for safeguarding data and information and increasing the level of legal action taken against breaches of modern technologies that lack legal protection. Previously, Senyo et al. (2016) reported that government legislation regarding modern technologies is critical to maintaining the security and transparency of innovative technologies such as blockchain or metaverse technologies. Government involvement can be crucial in adverse circumstances or in invoking force majeure. Nkhoma et al. (2013) showed that regulatory support comes in the form of legislation that protects and promotes the adoption of new technologies by organizations. As a result, regulatory support is adopted as a critical factor in this study that influences companies' adoption of blockchain technology.

Hypothesis 3. (H3): The environmental elements (i.e., rivalry pressure, business partner pressure, and regulatory support) of blockchain technology are related to its resistance.

2.5.4. Resistance to blockchains and blockchain strategies

Resistance to innovation can be categorized into two types: passive and active. Komulainen and Nätti (2023) describe passive resistance as a state of aversion to innovation caused by satisfaction with the status quo and an unwillingness to change, and it is not dependent on a specific technology or innovation. Conversely, Wong et al. (2023) characterized active resistance as a negative evaluation of a particular technology on the basis of antagonistic attitudes, behavior, perceptions, and intentions before adoption. Friedman and Ormiston (2022) and Katebi et al. (2022) showed that users' perceived difficulty in understanding or using technology is a significant barrier to technology adoption. This perception can produce value barriers when organizations fail to anticipate what could be improved if the technology is adopted (Kautish et al., 2023). Sadiq et al. (2021) reported that value barriers significantly contributed to consumer resistance to the adoption of eco-friendly cosmetics in India. Therefore, if an organization does not view innovation as more valuable than traditional methods do, then it will resist adoption. Friedman and Ormiston (2022) and Kleijnen et al. (2009) found that risk concerns play a central role in resistance to technology adoption when the technology is unreliable, uncertain, or performs unsatisfactorily. For example, as reported by Dehghani et al. (2022) and Dwivedi et al. (2023), when organizations perceive that an innovation is inappropriate or deficient due to a lack of necessary information about the innovation or a lack of awareness of its potential, this perception could lead to psychological risk barriers. Kumar et al. (2023) found that privacy concerns related to blockchain technology generate resistance to adoption. Given that blockchain is still a nascent technology and requires auxiliary tools for implementation, its widespread adoption is limited (Walsh et al., 2021). If users feel that blockchain stakeholders threaten their privacy and data security, then they will resist adopting blockchain (Dwivedi et al., 2023). Komulainen and Nätti (2023) found that the cost of switching from a traditional system to a blockchain can also be a factor that limits its adoption. In the findings of Jang et al. (2023) on the theory of resistance to innovation, people always seek to continue with the status quo even if there are potential alternatives and improvements.

Hypothesis 4. (H4): Resistance to blockchain technology is related to the blockchain strategy.

2.5.5. Resistance to the blockchain as a mediator

Friedman and Ormiston (2022) and Kouhizadeh et al. (2021) argued that although blockchain technology has garnered significant attention and popularity as a disruptive technology, it has prompted organizations to adopt it. Several promising features of blockchain technology have been proposed, including cost savings, enhanced traceability transparency, an improved product lifecycle, sustainability, and other benefits (Chan et al., 2023; Shahzad et al., 2024). However, perceived barriers to blockchain adoption are increasing (Kouhizadeh et al., 2021). Friedman and Ormiston (2022) reported that these barriers create negative attitudes among stakeholders, especially since investing in blockchain technology requires money, which is irretrievable. In the context of innovation adoption, Kautish et al. (2023) suggested that innovation is often risky because of stakeholders' ignorance or lack of awareness to assess how innovation can improve the current situation. Talwar et al. (2023) and Wong et al. (2023) found that stakeholders may face psychological or functional barriers that push them to resist blockchain due to insufficient information or fear of disrupting the status quo. Therefore, adopting blockchain technology first requires overcoming the barriers behind resistance. Previous studies have shown that alliances between stakeholders' networks in business models increase the chances of creating innovative ecosystems and regulatory contexts that support the implementation of new technologies. This scenario increases the potential opportunities for industries to be open to new ideas and technologies and for all stakeholders to participate in fostering a robust and innovative information system (Mention, 2011). Although individuals fear that their data will be hacked via blockchain technology, their positive perception of the consequences will reduce resistance to adoption. Khaw et al. (2023), Sadaa et al. (2023), and Wong et al. (2023) reported that data can also be considered a strategic resource for building profitable blockchain technology. On the basis of the discussions presented above, resistance to blockchain adoption by organizations can be reduced by controlling or reducing the functional and psychological barriers that oppose adoption.

Hypothesis 5. (H5): Resistance to blockchain technology mediates the relationship between the elements of TOE and the blockchain strategy.

3. Methodology

The main objectives of this study are to determine the impact of TOE-related factors and examine the resistance to blockchain technology adoption on the blockchain strategy in the context of Iraq's oil supply companies. Oil companies in Iraq are major contributors to the world's oil production. According to recent reports published in June 2024 by the Extractive Industry Transparency Initiative, the crude oil production in Iraq is 3029 barrels per day. Many multinational companies are involved in oil extraction and export activities. According to the data of the Ministry of Oil in Baghdad, Iraq oil reserves are expanded across 67 oil fields comprising 7 major fields. The major oil fields include Halfaya, Kirkuk, Majnoon, Rumaila, West Qurna 1 and 2, and the Zubair Oil Field. In addition, the oil sector in Iraq is supported by 27 international oil companies in addition to local companies. The research population comprises 32,059 individuals working across these oil companies. The survey includes employees working as managers, heads of departments, and divisional managers of oil companies. In accordance with the recommendations of Krejcie and Morgan (1970), the sample size was determined to be 381 employees, among whom the questionnaire was distributed. After several follow-ups, we received 313 valid responses. The following table presents the major demographics of the respondents. A five-point Likert scale (1: strongly disagree to 5: strongly agree) was used to obtain the response.

Previous studies in business research confirm that common method bias is a sensitive issue for single-source data. They affirm that academics should address common method bias by employing procedural and statistical measures (Alnoor et al., 2022). In terms of procedural measures, questionnaire items should be clarified, and respondents' confidentiality should be assured. In terms of statistical measures, Harman's single-factor method is the most suitable for addressing data bias. The results confirmed that the variance rates were less than 50 %, which is a bright sign that the data do not have common bias. The analysis section enumerates the items that were used to determine the proposed relationships. In this research, a hybrid structural equation modeling (SEM)–artificial neural network (ANN) methodological approach was adopted to investigate the noncompensatory and causal relationships among the variables. This hybrid approach is considered more reliable and leads to more reliable and theoretically and practically rigorous results. The hybrid SEM–ANN approach is also important for avoiding validity issues, as may be the case when a single SEM approach is used.

4. Results

4.1. PLS–SEM and ANN justification

Zaidan et al. (2023) found that social science researchers have relied for decades on a variety of methods to analyze data. In this context, statistical methods have been expanded from limited first-generation and second-generation techniques to analyze multivariate data. The first-generation methods include regression analysis, variance analysis, exploration techniques, confirmatory analysis, and logistic regression analysis. The second generation of techniques consists of SEM analyses. Given the inherent weakness in first-generation methods, the use of second-generation techniques has gained popularity. The SEM approach is divided into two types: the first type relies on covariance to reject and accept theories and is called covariance-based SEM (CB-SEM); the second type reveals the development of theories and explains the variance in the dependent variable and is called the partial least square (PLS)–SEM approach (Albahri et al., 2022). PLS–SEM analysis matches the objectives of this study to capture the causal relationships between TOE factors and blockchain technology. PLS–SEM, a widely accepted method in social and business research, was employed to capture linear relationships between variables. Thus, PLS–SEM is a robust approach for analyzing variance in multiple dependent variables. In addition, PLS–SEM offers predictive advantages such as the coefficient of determination (R2) (Alnoor et al., 2022).

The PLS–SEM approach has been widely used in many applications, and at the same time, a stream of new research has emerged that integrates the ANN approach with PLS–SEM. The emerging new statistical generations aim to provide nonlinear relationships and validate the results of second-generation methods (Zaidan et al., 2023). The ANN approach allows prediction of the influence of the antecedents of the dependent variable by capturing noncompensatory relationships; it complements PLS–SEM analysis and provides conclusive results for hypothesis testing, especially since the PLS–SEM approach suffers from bias in the results because of its reliance on a single-step approach to data collection (Alnoor et al., 2024c). ANN analysis also overcomes the problem of the cross-sectional design of data by presenting the contribution of each factor to the studied outcome. The use of ANN analysis with PLS–SEM has increased in the research stream for different types of aspects related to education, health, tourism, e-commerce, marketing and other business sectors (Albahri et al., 2022). Keeping the proven record of the success of combining PLS–SEM with ANN in prior research (e.g., Alnoor et al., 2024a; Khaw et al., 2022), linear and noncompensatory relationship approaches have been employed to provide fresh academic insight and rigorous methodological contributions to practitioners.

4.2. PLS–SEM analysis

A two-stage analysis—the first stage included the PLS–SEM approach, and the second stage involved ANN analysis—was applied in this work. As mentioned in previous studies, ANN analysis complements the second-generation PLS–SEM approach by capturing noncompensatory relationships and testing the validity of the PLS–SEM results. PLS–SEM analysis captures causal relationships between variables. Causal relationship analysis includes measurement model testing and structural model testing. Convergent validity and discriminant validity must be verified in the testing of a measurement model. Convergent validity involves factor loadings (FLs), construct reliability (CR), and average variance extracted (AVE). Table 2 illustrates the convergent validity test. (See Fig. 1.) (See Table 1.)

The results revealed that the FL values for all the items were greater than 0.70, whereas the CR and AVE values for the variables were greater than 0.70 and 0.50, respectively. The results confirm that convergent validity is not a concern. However, discriminating against validity is an issue that cannot be overlooked. Three methods, namely, the Fornell–Larcker criterion, cross-loadings, and the heterotrait–monotrait (HTMT) ratio of correlations, can be used to test discriminant validity. The HTMT method is widely used, accepted, and recommended in the literature. The HTMT method was used to test discriminant validity. Table 3 presents the results of the discriminant validity test.

The results of the correlation matrix for the HTMT test revealed that the correlation values between the variables were less than 0.90 and 0.85. These results provide insight into the absence of concerns regarding discrimination validity. As mentioned earlier, the PLS–SEM approach involves testing the structural model. Therefore, bootstrapping and resampling were applied in this research by adopting 5000 resamples and bias-corrected confidence intervals (one-tailed, 0.05). Table 4 presents the results of the hypothesis testing.

The results of the hypothesis testing indicated that all the hypotheses were accepted. The results reveal that TOE factors play a significant role in the implementation and adoption of the blockchain strategy. In other words, positive relational influences exist among relative advantage, compatibility, trust, security, lack of technological knowledge, firm size, rivalry pressure, business partner pressure, and regulatory support for the blockchain strategy. In addition, resistance to blockchain mediates the relationships among technological, organizational, and environmental factors and blockchain strategy.

Table 2
Convergent validity.

Variables	Items	LF	CR	AVE	Source
Relative advantage	Using blockchain technology assists to achieve undertakings more rapidly.	0.870	0.909	0.714	Chittipaka et al. (2023)
	Using blockchain can empower us to reduce costs.	0.892			
	Applying blockchain administration increases productivity.	0.821			
	Employing blockchain machinery can increase adaptability.	0.793			
Compatibility	The resemblance of blockchain with the work would urge us to use it.	0.846	0.913	0.726	Chittipaka et al. (2023)
	The blockchain used is compatible with enterprise requirements.	0.905			
	The blockchain technology used fits our business style.	0.739			
	The blockchain used does not fit the business requirements.	0.907			
Trust	I feel confident placing away my business's information operating blockchain.	0.745	0.748	0.760	Chittipaka et al. (2023)
	Placing away our business's information under unknown operate is one of our concerns.	0.749			
	There is a possibility that my business's information may be used in the cloud without consent.	0.747			
Security	I believe that blockchain isn't sufficiently confident to store our business's information.	0.799	0.858	0.603	Chittipaka et al. (2023)
	I believe that traditional machineries are dangerous than blockchain technology.	0.762			
	Security factor affects the privacy and use of blockchain technology.	0.795			
	I believe that the risks of blockchain technology that overdo the aids.	0.780			
Lack of technological knowledge	Employees have little knowledge of how to use blockchain technology.	0.927	0.949	0.861	Dehghani et al. (2022)
	Employees do not have the technical knowledge to use blockchain technology.	0.984			
	Technology experts report a lack of knowledge in understanding blockchain.	0.870			
Firm size	Larger firm is more benefited from blockchain.	0.719	0.809	0.796	Premkumar and Roberts (1999)
	Smaller firms can use pay-per-use service.	0.739			
	Smaller firms avoid using blockchain technology.	0.764			
Rivalry pressure	The approved blockchain technology provides accuracy.	0.854	0.944	0.809	Dehghani et al. (2022)
	Blockchain provides better services that increase the efficiency of the company.	0.748			
	The blockchain used provides a better understanding of the competitiveness of the market.	0.826			
	Blockchain helps in maintaining authority in the process of data sharing.	0.865			
Business partners pressure	Using blockchain can enhance matching among business partners.	0.877	0.93	0.769	Lin and Lin (2008)
	Business partners recommend adopting blockchain technology in my company.	0.853			
	Adopting blockchain technology increases the trust of business partners.	0.915			
	Business partners in my company oppose adopting blockchain technology.	0.862			
	Business partners support the company's decision to use blockchain.	0.736			
	Our business partners do not advocate the company's decision to use blockchain.	0.803			
Regulatory support	My company supports using blockchain technology.	0.899	0.846	0.587	Alain and Ooi (2008)
	My company provides mechanical guidelines for using blockchain technology.	0.863			
	Existing laws and regulations are sufficient to ensure that data is stored in virtual storage.	0.743			
	The virtual resource guidelines are in compliance with the law in my company.	0.707			
Resistance toward Blockchain	I will oppose modifying the systems used to systems based on blockchain technology.	0.737	0.901	0.695	Walsh et al. (2021)
	I do not agree to switching the systems used to systems based on blockchain technology.	0.842			
	I will not cooperate with switching the systems used to systems based on blockchain technology.	0.902			
	I will not comply with switching some of the systems used to systems based on blockchain technology.	0.847			
Blockchain strategy	My company is adopting a strategy to invest resources in the blockchain supported supply chain.	0.773	0.837	0.632	Adel and Younis (2023)
	My company's blockchain strategy business activities require the use of blockchain.	0.799			
	The functional areas for adopting a blockchain strategy in my company demand the employ of blockchain.	0.812			

Table 3
Discriminant validity.

Variables	1	2	3	4	5	6	7	8	9	10	11
1. Blockchain strategy											
2. Business pressure	0.169										
3. Compatibility	0.646	0.075									
4. Firm size	0.141	0.122	0.063								
5. Lack of knowledge	0.108	0.036	0.035	0.110							
6. Regulatory support	0.348	0.128	0.337	0.056	0.081						
7. Relative advantage	0.106	0.033	0.050	0.128	0.042	0.121					
8. Resistance to blockchain	0.377	0.045	0.276	0.069	0.054	0.183	0.094				
9. Rivalry pressure	0.141	0.078	0.070	0.051	0.079	0.932	0.110	0.056			
10. Security	0.202	0.071	0.264	0.068	0.041	0.100	0.101	0.110	0.054		
11. Trust	0.239	0.055	0.096	0.049	0.051	0.097	0.318	0.074	0.114	0.05	

4.3. ANN analysis

The combination of PLS–SEM and ANN yields a promising method for predicting the determinants of the blockchain strategy. The combination of PLS–SEM analysis tests the theories of linear models and relationships between different constructs of the proposed model (Chew et al., 2023). However, nonlinear relationships in the second-generation approach are still overlooked. The ANN method can detect non-compensatory relationships and determine the importance and contribution of each determinant to the blockchain strategy. In addition, the

Table 4
Hypothesis testing.

Direct path				
Path	O	STDEV	T	P
Relative advantage - > Resistance to blockchain	0.171	0.072	2.375	0.000
Compatibility - > Resistance to blockchain	0.233	0.054	4.315	0.020
Trust - > Resistance to blockchain	0.174	0.033	5.273	0.000
Security - > Resistance to blockchain	0.153	0.075	2.040	0.000
Lack of knowledge - > Resistance to blockchain	0.119	0.024	4.958	0.000
Firm size - > Resistance to blockchain	0.193	0.018	10.722	0.000
Rivalry pressure - > Resistance to blockchain	0.165	0.034	4.853	0.010
Business pressure - > Resistance to blockchain	0.101	0.023	4.391	0.000
Regulatory support - > Resistance to blockchain	0.144	0.034	4.235	0.000
Resistance to blockchain - > Blockchain strategy	0.323	0.051	6.333	0.006
Indirect path				
Relative advantage - > Blockchain strategy	0.223	0.024	9.292	0.000
Compatibility - > Blockchain strategy	0.275	0.024	11.458	0.002
Trust - > Blockchain strategy	0.224	0.034	6.588	0.000
Security - > Blockchain strategy	0.217	0.025	8.680	0.000
Lack of knowledge - > Blockchain strategy	0.106	0.021	5.048	0.000
Firm size - > Blockchain strategy	0.130	0.035	3.714	0.000
Rivalry pressure - > Blockchain strategy	0.121	0.031	3.903	0.000
Business pressure - > Blockchain strategy	0.201	0.024	8.375	0.000
Regulatory support - > Blockchain strategy	0.246	0.032	7.688	0.003

Table 5
Root mean square of error (RMSE) values.

ANN	Input neurons: TOE factors						Total
	Output nodes: Blockchain strategy						
	Training			Testing			
	N	SSE	RMSE	N	SSE	RMSE	
1	273	750.185	0.706	40	90.170	0.700	313
2	275	625.177	0.600	38	80.745	0.711	313
3	276	596.732	0.585	37	69.215	0.541	313
4	270	741.643	0.697	43	101.804	0.674	313
5	264	722.246	0.691	49	103.730	0.653	313
6	278	793.862	0.701	35	77.011	0.629	313
7	272	727.964	0.701	41	92.838	0.707	313
8	276	803.208	0.717	37	83.709	0.595	313
9	268	732.074	0.683	45	95.517	0.667	313
10	276	712.219	0.667	37	80.952	0.676	313
Mean		720.531	0.675		87.569	0.655	
SD		65.164	0.046		11.131	0.054	

ANN approach validates the results of PLS–SEM (XinYing et al., 2024). The ANN method functions, such as the human brain, and consists of neural networks that store information and knowledge and allow them to be used and reused. Neural networks stock information resulting from learning processes and store it in neurons (Abdullah et al., 2023). In line with the work of Alnoor et al. (2024b), a feed-forward-back-propagation algorithm with multiple layers was used. Fig. 2 shows the ANN model. The ANN model consists of three layers, namely, hidden layers, input layers, and output layers. Small multilayer perceptions are decision-making units with each layer supporting data transformations to make predictions that help determine the contribution of TOE factors to the blockchain. Data are processed via multiple units or layers, and each layer employs several calculations. Although activation functions, specifically the sigmoid function, help produce values ranging from zero to one in the perceptual layers, the sigmoid function can help in making decisions and predictions (Chew et al., 2023). Multiple layers also interpret complex relationships and fathom input–output relationships. Multiple layers and the sigmoid function collectively contribute to high prediction accuracy regardless of the type of data used. In this context, the sigmoid function is important for predicting the probability of a certain outcome. Moreover, the combination of multiple layers and a sigmoid function is crucial in rational decision-making processes (Kurani et al., 2023).

ANNs and PLS–SEM complement each other by taking advantage of linear and nonlinear features. The black box function of ANN analysis contributes to predicting the determinants of the blockchain strategy but fails to test hypotheses. Sigmoid functions were used in this research for the activation functions and hidden layers. The extant literature recommends the use of sigmoid functions because they facilitate the understanding of complex models and nonlinear activation (Leong et al., 2020). In the ANN model, a tenfold approach was used to increase the generalizability of the results and provide predictive power and robustness to the ANN model. In addition, tenfold sampling allows the sample to be divided into subsamples that permit training and testing the model on different parts of the data. This approach provides a reliable estimate of the model's performance by making the most of the data; it also helps reduce the bias in the subsets of data by using 10 % of the data for testing and 90 % of the data for training to make informed decisions by choosing a better model. The approach also significantly reduces random variations in datasets that are divided into multiple samples (Chew et al., 2023; Khaw et al., 2022) and helps reduce the overfitting problem. Finally, such an approach improves the performance of the model to work well on training data and unseen data. For example, providing an ANN model with data related to a certain phenomenon helps train the model to understand the phenomenon. However, when circumstances change or new developments emerge that alter the form of the phenomenon under study, the use of the tenfold approach increases the understanding of the new phenomenon. The tenfold method splits the data into ten sets, the model is trained nine times, and the remaining set is tested. Each time, a different set is tested to provide high prediction accuracy, and the model is trained for different processes and conditions that affect the prediction (Bejani and Ghatee, 2021). The significant paths of PLS–SEM analysis are the inputs to the ANN model. Thus, the ANN model is constructed on the basis of the variables supported by the PLS–SEM method. Table 5 shows the analysis of the ANN model.

The RMSE results indicate the predictive accuracy of the ANN model. The average values for training and testing were 0.675 and 0.655, respectively. The mean square error results reveal that the RMSE values were less than one and were close to one another. Moreover, the RMSE results and values provide insight into the predictive accuracy of the ANN model. ANN results, which capture the importance of variables, were used to determine the contribution of TOE factors to the blockchain strategy. Table 6 presents the sensitivity analysis of the ANN results.

The ANN results highlight the importance of TOE factors in implementing a blockchain strategy and mitigating resistance to the use of more advanced technologies. The sensitivity analysis results confirm that the blockchain strategy is the most sensitive to relative advantage, followed by rivalry pressure, trust, security, lack of technological knowledge, and regulatory support. The prediction accuracy was determined by measuring the goodness-of-fit index, i.e., the RMSE was divided by the sum square of the error of the test results. The results confirmed that the goodness-of-fit index was 99 %. Cross-validation is used to assess the reliability and the lack of prejudice in dataset 313. The reliability of the dataset could be checked via tenfold cross-validation, in which the ANN model could be applied for each fold. In these steps, the dataset is equally split into ten subsets, nine of which are used for training and one for validation. The process is repeated ten times with distinct sets of data to be used as a validation set each time. If the performance of the model is quite similar across all the folds, then the data are consistent, and the model is not biased toward a particular part of the data. The consistency of the results across all folds implies the absence of

Table 6
Sensitivity analysis.

ANN	Relative importance									
	REA	COM	TRS	SEC	LTK	FIZ	RIP	BPP	RES	RTB
1	0.105	0.200	0.078	0.095	0.045	0.098	0.075	0.091	0.119	0.094
2	0.116	0.098	0.050	0.112	0.076	0.072	0.135	0.083	0.103	0.157
3	0.117	0.099	0.090	0.101	0.060	0.082	0.088	0.083	0.137	0.142
4	0.114	0.132	0.103	0.087	0.047	0.064	0.119	0.067	0.074	0.193
5	0.080	0.103	0.101	0.157	0.065	0.062	0.121	0.054	0.073	0.185
6	0.132	0.112	0.066	0.076	0.072	0.083	0.153	0.065	0.111	0.129
7	0.109	0.126	0.075	0.104	0.062	0.051	0.149	0.091	0.088	0.145
8	0.086	0.110	0.116	0.087	0.080	0.048	0.174	0.083	0.099	0.117
9	0.109	0.138	0.085	0.114	0.072	0.075	0.075	0.090	0.058	0.184
10	0.141	0.138	0.072	0.095	0.078	0.074	0.113	0.071	0.107	0.110
Mean importance	0.145	0.048	0.130	0.126	0.126	0.066	0.134	0.080	0.115	0.031
Importance (%)	100.0 %	32.9 %	89.8 %	86.7 %	86.6 %	45.2 %	92.6 %	55.0 %	79.4 %	21.3 %

Notice: REA = Relative advantage, COM = Compatibility, TRS = Trust, SEC = Security, LTK = Lack of technological knowledge, FIZ = Firm size, RIP = Rivalry pressure, BPP = Business partners pressure, RES = Regulatory support, RTB = Resistance toward blockchain.

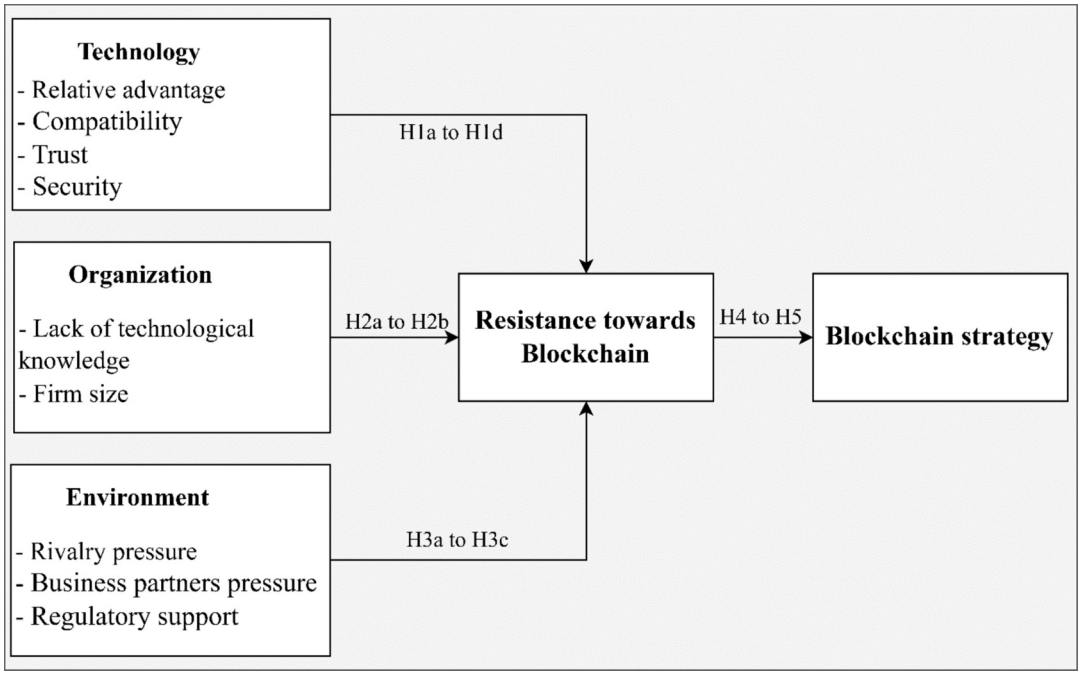


Fig. 1. Conceptual framework.

Table 1
Demographics of respondents.

Demographics		Percentage (%age)
Gender	Males	79
	Females	21
Age	20 years – 30 years	19
	31 years – 40 years	35
	41 years – 50 years	29
	51 years – 60 years	17
Education	Bachelor's degree holders	38
	Master's degree holders	32
	Diploma holders	24
	Doctoral degree holders	6
Work Experience	Less than 10 years	16
	11 years – 20 years	13
	21 years – 30 years	29
	31 years and beyond	42

major variations or outliers in the dataset used; therefore, the dataset can be used in more extensive analysis or modeling. Fig. 3 shows the tenfold cross-validation.

The results of each fold were quite similar and nearly identical to each other; this is evidenced by the accuracy being similar for all the folds. Therefore, the dataset is balanced without bias. This validation step ensures that the dataset used in this study can be relied on for analysis and modeling.

5. Discussion

The determinants of blockchain strategy were evaluated in this study from the perspective of the elements of TOE theory in the context of oil companies working in Iraq. The mediating role of resistance to blockchain technology was also investigated. The proposed relationship was tested in this study by presenting five hypotheses. Evidence in support of the proposed hypotheses was established. With respect to the first hypothesis, the relative advantage of blockchain technology is that it reduces resistance to blockchain and thus is negatively related to resistance to blockchain. These findings are aligned with earlier studies advocating that blockchain technology reduces costs and increases productivity by generating less resistance and greater adaptability of blockchain (Chittipaka et al., 2023). Similarly, Chittipaka et al. (2023)



Fig. 2. ANN model.

and Ghode et al. (2020) argued that relative advantage is one of the critical factors in the adoption of a blockchain strategy. Yuen et al. (2021) argued that, compared with other technologies, blockchain is more reliable in terms of information accuracy and reliability. In addition, Kamble et al. (2021) found that the relative advantage of blockchain increases the perceived usefulness and ease of application. Thus, the relative advantage of blockchain technology is significantly and negatively correlated with resistance to its application. Furthermore, the compatibility of blockchain technology is negatively correlated with its resistance. In this context, the findings of this research support earlier studies showing that compatible technologies are easier to adopt and implement (Alnoor et al., 2024b; Kamble et al., 2021; Oliveira et al., 2016). Compatibility enables the organization to integrate the new technology into the extant system and thereby mitigates resistance to

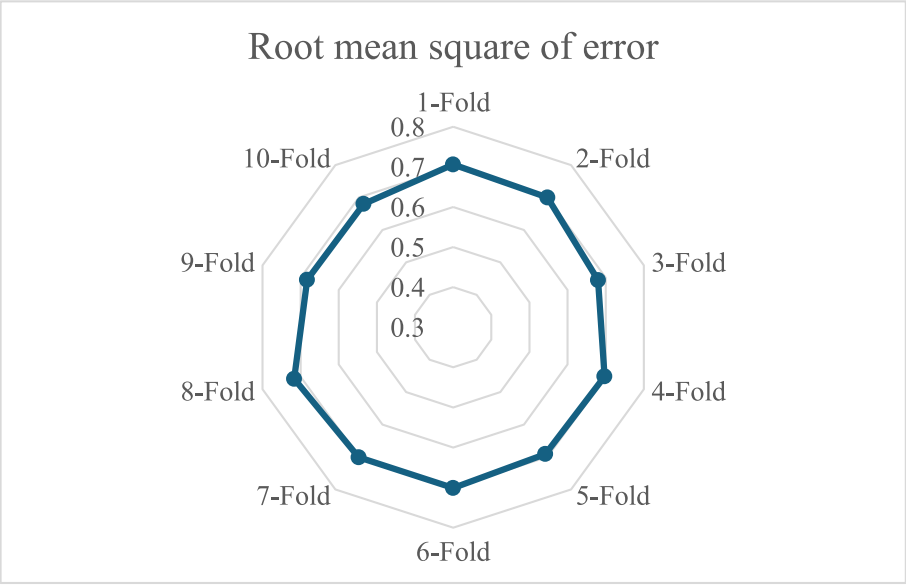


Fig. 3. Tenfold cross validation.

new technology (Dutta et al., 2020; Shahzad et al., 2024). The findings revealed that trust in blockchain is negatively related to resistance to blockchain technology. For business firms, blockchain facilitates the traceability and transparency of business transactions, which increases trust in blockchain (Chew et al., 2024; Montecchi et al., 2019). Garaus and Treiblmaier (2021) consider trust as another prominent factor in adopting blockchain technology; the higher the level of trust in blockchain is, the lower the resistance to its adoption. The ability of the security features of blockchain technology to reduce resistance to its adaptation was also hypothesized, as organizations are more concerned about the privacy and security of consumers' data and other information. As demonstrated by Shahzad et al. (2024) and Tseng et al. (2023), organizations are reluctant to adopt technologies that compromise security features. Nonetheless, if the security features of blockchain help organizations avoid online risk and fraud, then they are likely to readily accept blockchain technology (Behnke and Janssen, 2020; Kouhizadeh et al., 2021).

The second aspect of TOE relates to the organizational elements, which include lack of technological knowledge and firm size (as far as this study is concerned). The lack of technological knowledge and resistance to blockchain are hypothesized to be positively related. The findings of this research corroborate earlier studies showing that a lack of relevant skills and expertise hinders organizations from adopting new technologies and vice versa (Dehghani et al., 2022; Kaur et al., 2021). Similarly, enhanced levels of requisite skills and knowledge incline organizations to adopt and implement new technologies (Chen et al., 2022). Support is provided for the hypothesis proposed in this work that the larger the firm size is, the lower the resistance to blockchain technology. This situation occurs because larger firms are more diverse in terms of their technical knowledge and human abilities, which help them easily align with and adopt new technology (Shahzad et al., 2024). Given their sufficient resources, large organizations are more likely to adopt new technology than small organizations are (Clohessy and Acton, 2019).

The third aspect that has been investigated with respect to resistance to blockchain technology is the environment, which includes three elements: rivalry pressure, business partner pressure, and regulatory support. The findings provide evidence that rivalry pressure is negatively related to resistance to blockchain technology; this can be attributed to organizations' need to compete in the market to achieve their market share to respond to competitors' market tactics. When competitors adopt new technologies and introduce market innovations, they motivate organizations to adopt new technologies and implement innovations (Han and Chen, 2021; Thompson and Rust, 2023). These findings coincide with the works of Kamble et al. (2021) and Kouhizadeh et al. (2021), who suggested that rivalry pressure urges the adoption of new technologies and hence reduces resistance to blockchain technology. Regarding the other two elements, i.e., business partner pressure and regulatory support, the findings suggest that both are negatively correlated with resistance to blockchain technology. Resource contribution from the partner facilitates the acquisition of new technology. Similarly, pressure from a partner to achieve competitive advantage works in favor of new technology adoption (Baral and Verma, 2021; Francisco and Swanson, 2018). Thus, the notion that partner pressure is negatively related to resistance to blockchain technology is affirmed. Similarly, regulatory support reduces resistance and encourages organizations to adopt new technologies. On this basis, the findings of this research support the earlier studies of Chittipaka et al. (2023) and Baral and Verma (2021), who reported that regulatory support is essential to help organizations adopt new technologies.

Two hypotheses concerning the role of resistance to blockchain technology were also tested in this study. First, resistance to blockchain is related to the blockchain strategy, and second, resistance to blockchain mediates the relationship between the elements of TOE (as predictors) and the blockchain strategy. First, the findings of this research support the hypothesis that resistance to blockchain is negatively related to the blockchain strategy. Privacy and security issues cause resistance, which negatively affects the blockchain strategy. Second, the technology has not reached its maturity, and concerns about switching from existing technology to novel technology abound. This issue underscores the level of resistance, further suggesting negative consequences for the blockchain strategy (Dwivedi et al., 2023; Walsh et al., 2021). Moreover, sufficient evidence is provided for the mediating role of resistance to blockchain in the relationship between TOE elements and the blockchain strategy. Hence, all the hypotheses in this research are acceptable.

Theoretical and Practical Implications.

Some important theoretical and practical implications can be elucidated from this research. First, the study was conducted in a relatively novel context, which enriches the understanding of the relationship between the elements of TOE and the blockchain strategy. Second, aspects of TOE, which were largely overlooked in earlier studies, were considered. The researchers of this work argue that, given the absence of

technical knowledge and regulatory support, blockchain technology may face more resistance, thereby influencing oil companies' blockchain strategy. Researchers should understand that oil companies in Iraq are operating in a turbulent environment where security and the absence of appropriate regulations are deep concerns. Devising a blockchain strategy seems to be a challenging task for oil companies. Furthermore, the way resistance is developed toward blockchain technology and its implications for the development of a blockchain strategy are unearthed. Understanding the mediating role of resistance and the factors that cause resistance is important. This study provides sufficient evidence to this end. Methodologically, this work implies a synergetic PLS–SEM and ANN approach to obtain more robust results. Previous studies have relied mostly on a single methodological approach. Thus, the results of this work are more grounded, robust, and reliable. Practically, some crucial elements of TOE can be considered important for managers of oil companies to understand the nature of resistance to blockchain technology and for the development of an appropriate blockchain strategy. The findings can help oil companies' managers devise relevant blockchain strategies while considering the importance of the TOE framework. The managers of oil companies need to pay more attention to TOE-related factors to transition the existing system toward the adoption of blockchain.

6. Avenues for future research

A significant future research agenda was unearthed in this work. As discussed earlier, blockchain technology is rapidly replacing traditional technologies, and a similar nature of study in other contexts, i.e., oil companies operating in other parts of the world, is proposed. This aspect will provide a comparison between our studies and other works in terms of the results and implications. The mediating role of resistance was also discussed; on this basis, the moderating role of resistance to blockchain technology may be determined. Furthermore, future studies can elaborate on other aspects of the TOE framework, such as the perceived value of technology, technological infrastructure, organizations' subjective norms, organizations' demographic composition, the scope of organizations' business, external support, and trading partners' readiness to embrace change. Finally, data across different organizational levels must be collected to determine the level of readiness/resistance to the blockchain.

7. Conclusion

The relationships among TOE elements, resistance to blockchain technology, and the blockchain strategy were investigated in this work. In addition to the direct relationship between the mentioned variables, the mediating role of resistance to blockchain technology was investigated. The context of oil companies operating in Iraq was considered. The purpose of choosing oil companies was to recognize their role in the economic development of Iraq. Several hypotheses were used on the basis of a comprehensive review of the relevant literature to investigate the proposed relationship. Data were collected from manager-level employees through a questionnaire. The PLS–SEM and ANN approaches were employed to analyze the data and determine the proposed relationships. All the hypotheses were supported and consequently accepted. TOE-related elements significantly influence resistance to blockchain technology and consequently affect the blockchain strategy. The essential conclusion of this study is that blockchain is swiftly overtaking existing technologies, and it is pertinent that oil companies should be more vigilant about the factors that are important to the adoption of blockchain technology.

CRediT authorship contribution statement

Alhamzah Alnoor: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation. **Sammar Abbas:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology. **Abdullah Mohammed Sadaa:** Writing – review & editing, Validation, Project administration, Methodology, Data curation. **XinYing Chew:** Writing – review & editing, Visualization, Validation, Supervision, Software, Funding acquisition. **Gül Erkol Bayram:** Writing – review & editing, Resources, Methodology, Funding acquisition, Data curation, Conceptualization.

Author contribution

All authors contributed to the study's conception and design. Material preparation, data collection, and analysis were performed by all authors. All authors read and approved the final manuscript.

Data availability

Data will be made available on request.

References

Abdullah, H.O., Atshan, N., Al-Abrrow, H., Alnoor, A., Valeri, M., Erkol Bayram, G., 2023. Leadership styles and sustainable organizational energy in family business: modeling non-compensatory and nonlinear relationships. Journal of Family Business Management 13 (4), 1104–1131.

Adel, H.M., Younis, R.A.A., 2023. Interplay among blockchain technology adoption strategy, e-supply chain management diffusion, entrepreneurial orientation and human resources information system in banking. Int. J. Emerg. Mark. 18 (10), 3588–3615.

Ahmad, R.W., Salah, K., Jayaraman, R., Yaqoob, I., Omar, M., 2022. Blockchain in oil and gas industry: applications, challenges, and future trends. Technol. Soc. 68 (2), 1–15.

Ahmadi, H., Nilashi, M., Shahmoradi, L., Ibrahim, O., 2017. Hospital information system adoption: expert perspectives on an adoption framework for Malaysian public hospitals. Comput. Hum. Behav. 67 (2), 161–189.

Ajzen, I., 1985. From intentions to actions: A theory of planned behavior. In: Action Control: From Cognition to Behavior. Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 11–39.

Alain, C.Y.L., Ooi, K.B., 2008. Adoption of interorganizational system standards in supply chains: an empirical analysis of RosettaNet standards. Ind. Manag. Data Syst. 108 (4), 529–547.

Alalwan, A.A., 2020. Mobile food ordering apps: an empirical study of the factors affecting customer e-satisfaction and continued intention to reuse. Int. J. Inf. Manag. 50 (5), 28–44.

Albahri, A.S., Alnoor, A., Zaidan, A.A., Albahri, O.S., Hameed, H., Zaidan, B.B., Yass, A. A., 2022. Hybrid artificial neural network and structural equation modelling techniques: a survey. Complex Intell. Syst. 8 (2), 1781–1801.

Alharbi, F., Atkins, A., Stanier, C., 2016. Understanding the determinants of cloud computing adoption in Saudi healthcare organisations. Complex Intell. Syst. 2 (1), 155–171.

Alnoor, A., Al-Abrrow, H., Al Halbusi, H., Khaw, K.W., Chew, X., Al-Maatoq, M., Alharbi, R.K., 2022. Uncovering the antecedents of trust in social commerce: an application of the non-linear artificial neural network approach. Competitiveness Review: An International Business Journal 32 (3), 492–523.

Alnoor, A., Abbas, S., Khaw, K.W., Muhsen, Y.R., Chew, X., 2024a. Unveiling the optimal configuration of impulsive buying behavior using fuzzy set qualitative comparative analysis and multi-criteria decision approach. J. Retail. Consum. Serv. 81 (11), 1–17.

Alnoor, A., Atiyah, A.G., Abbas, S., 2024b. Blockchain adoption strategy from the perspective of organization, technology and environment theory: a hybrid SEM and ANN approach. Glob. Bus. Rev. (Article in press, 1-24).

Alnoor, A., Tiberius, V., Atiyah, A. G., Khaw, K. W., Yin, T. S., Chew, X., & Abbas, S. (2024c). How positive and negative electronic word of mouth (eWOM) affects customers' intention to use social commerce? A dual-stage multi group-SEM and ANN analysis. International Journal of Human–Computer Interaction, 40(3), 808–837.

Angelis, J., Da Silva, E.R., 2019. Blockchain adoption: a value driver perspective. Bus. Horiz. 62 (3), 307–314.

Antioco, M., Kleijnen, M., 2010. Consumer adoption of technological innovations: effects of psychological and functional barriers in a lack of content versus a presence of content situation. Eur. J. Mark. 44 (11/12), 1700–1724.

Anwar, A., Thongpapanl, N., Ashraf, A.R., 2021. Strategic imperatives of mobile commerce in developing countries: the influence of consumer innovativeness, ubiquity, perceived value, risk, and cost on usage. J. Strateg. Mark. 29 (8), 722–742.

Azzi, R., Chamoun, R.K., Sokhn, M., 2019. The power of a blockchain-based supply chain. Comput. Ind. Eng. 135 (9), 582–592.

Babich, V., Hilary, G., 2019. Blockchain and other distributed ledger technologies in operations. Found. Trends Technol. Info. Oper. Manage. 12 (2-3), 152–172.

Baral, M.M., Verma, A., 2021. Cloud computing adoption for healthcare: an empirical study using SEM approach. FIIB Business Review 10 (3), 255–275.

Behnke, K., Janssen, M.F.W.H.A., 2020. Boundary conditions for traceability in food supply chains using blockchain technology. Int. J. Inf. Manag. 52 (6), 1–10.

Bejani, M.M., Ghatee, M., 2021. A systematic review on overfitting control in shallow and deep neural networks. Artif. Intell. Rev. 54 (8), 6391–6438.
Chan, H.L., Choi, T.M., Mendez De la Torre, D., 2023. The "SMARTER" framework and real application cases of blockchain. Technol. Forecast. Soc. Chang. 196 (11), 1–12.
Chang, S.E., Chen, Y., 2020. When blockchain meets supply chain: a systematic literature review on current development and potential applications. IEEE Access 8 (1), 62478–62494.
Chen, C.C., Chang, C.H., Hsiao, K.L., 2022. Exploring the factors of using mobile ticketing applications: perspectives from innovation resistance theory. J. Retail. Consum. Serv. 67 (3), 1–10.
Chew, X., Khaw, K.W., Alnoor, A., Ferasso, M., Al Halbusi, H., Muhsen, Y.R., 2023. Circular economy of medical waste: novel intelligent medical waste management framework based on extension linear Diophantine fuzzy FDOSM and neural network approach. Environ. Sci. Pollut. Res. 30 (21), 60473–60499.
Chew, X., Alnoor, A., Khaw, K.W., Sadaa, A.M., Al Halbusi, H., Muhsen, Y.R., 2024. Symmetric and asymmetric modeling to boost customers' trustworthiness in livestreaming commerce. Curr. Psychol. 1 (1), 1–19.
Chittipaka, V., Kumar, S., Sivarajah, U., Bowden, J.L.H., Baral, M.M., 2023. Blockchain Technology for Supply Chains operating in emerging markets: an empirical examination of technology-organization-environment (TOE) framework. Ann. Oper. Res. 327 (1), 465–492.
Choi, D., Chung, C.Y., Seyha, T., Young, J., 2020. Factors affecting organizations' resistance to the adoption of blockchain technology in supply networks. Sustainability (Switzerland) 12 (21), 1–37.
Clohessy, T., Acton, T., 2019. Investigating the influence of organizational factors on blockchain adoption: an innovation theory perspective. Ind. Manag. Data Syst. 119 (7), 1457–1491.
Cole, R., Stevenson, M., Aitken, J., 2019. Blockchain technology: implications for operations and supply chain management. Supply chain management: An international journal 24 (4), 469–483.
Conoscenti, M., Vetro, A., De Martin, J.C., 2017. Peer to peer for privacy and decentralization in the internet of things. In: 2017 IEEE/ACM 39th International Conference on Software Engineering Companion (ICSE-C), pp. 288–290.
Davis, F.D., 1989. Perceived usefulness, perceived ease of use, and user acceptance of information technology. MIS Quarterly: Management Information Systems 13 (3), 319–340.
Dehghani, M., William Kennedy, R., Mashatan, A., Rese, A., Karavidas, D., 2022. High interest, low adoption. A mixed-method investigation into the factors influencing organisational adoption of blockchain technology. J. Bus. Res. 149 (4), 393–411.
Dutta, P., Choi, T.-M., Somani, S., Butala, R., 2020. Blockchain technology in supply chain operations: applications, challenges and research opportunities. Transportation Research Part e: Logistics and Transportation Review 142 (10), 1–33.
Dwivedi, Y.K., Balakrishnan, J., Das, R., Dutot, V., 2023. Resistance to innovation: a dynamic capability model based enquiry into retailers' resistance to blockchain adaptation. J. Bus. Res. 157 (3), 1–15.
Fawcett, S.E., Magnan, G.M., McCarter, M.W., 2008. Benefits, barriers, and bridges to effective supply chain management. Supply Chain Management: An International Journal 13 (1), 35–48.
Francisco, K., Swanson, D., 2018. The supply chain has no clothes: technology adoption of Blockchain for supply chain transparency. Logistics 2 (1), 1–13.
Friedman, N., Ormiston, J., 2022. Blockchain as a sustainability-oriented innovation?: opportunities for and resistance to Blockchain technology as a driver of sustainability in global food supply chains. Technol. Forecast. Soc. Chang. 175 (2), 1–17.
Garaus, M., Treiblmaier, H., 2021. The influence of blockchain-based food traceability on retailer choice: the mediating role of trust. Food Control 129 (1), 1–12.
Ghode, D., Yadav, V., Jain, R., Soni, G., 2020. Adoption of blockchain in supply chain: an analysis of influencing factors. J. Enterp. Inf. Manag. 33 (3), 437–456.
Goel, A., Neduncheliyan, S., 2023. An intelligent blockchain strategy for decentralised healthcare framework. Peer Peer Netw Appl 16 (2), 846–857.
Gökalp, E., Coban, S., & Gökalp, M. O. (2019). Acceptance of blockchain based supply chain management system: research model proposal. 2019 1st international informatics and software engineering conference (UBMYK), 1–6.
Gong, B., Zhang, H., Gao, Y., Liu, Z., 2023. Blockchain adoption and channel selection strategies in a competitive remanufacturing supply chain. Comput. Ind. Eng. 175 (1), 1–20.
Gupta, S., Ghardallou, W., Pandey, D.K., Sahu, G.P., 2022. Artificial intelligence adoption in the insurance industry: evidence using the technology–organization–environment framework. Res. Int. Bus. Financ. 63 (12), 1–15.
Hackius, N., & Petersen, M. (2017). Blockchain in logistics and supply chain: trick or treat?. In digitalization in supply chain management and logistics: smart and digital solutions for an industry 4.0 environment. Proceedings of the Hamburg international conference of logistics (HICL), Vol. 23 (pp. 3-18). Berlin: epubli GmbH.
Han, M.S., Chen, W., 2021. Determinants of eco-innovation adoption of small and medium enterprises: an empirical analysis in Myanmar. Technol. Forecast. Soc. Chang. 173 (12), 1–9.
Hew, J.J., Leong, L.Y., Tan, G.W.H., Ooi, K.B., Lee, V.H., 2019. The age of mobile social commerce: An Artificial Neural Network analysis on its resistances. Technol. Forecast. Soc. Chang. 144, 311–324.
Huang, D., Jin, X., Coghlan, A., 2021. Advances in consumer innovation resistance research: a review and research agenda. Technol. Forecast. Soc. Chang. 166 (5), 1–21.
Iansiti, M., Lakhani, K.R., 2017. The truth about blockchain. Harv. Bus. Rev. 95 (1), 118–127.
Iyanna, S., Kaur, P., Ractham, P., Talwar, S., Islam, A.K.M.N., 2022. Digital transformation of healthcare sector. What is impeding adoption and continued usage of technology-driven innovations by end-users? J. Bus. Res. 153 (12), 150–161.
Jang, H., Song, J., Kim, S., Byun, J.H., Lee, K.G., Park, K.H., Kang, T., 2023. ANCA: artificial nucleic acid circuit with argonaute protein for one-step isothermal detection of antibiotic-resistant bacteria. Nat. Comm. 14 (1), 1–13.
Jasperson, Jon, Carter, P.E., Zmud, R.W., 2005. A comprehensive conceptualization of post-adoptive behaviors associated with information technology enabled work systems. MIS Q. 29 (3), 525–557.
Joachim, V., Spieth, P., Heidenreich, S., 2018. Active innovation resistance: an empirical study on functional and psychological barriers to innovation adoption in different contexts. Ind. Mark. Manag. 71 (5), 95–107.
Kabra, G., 2023. Determinants of blockchain adoption and organizational performance in the healthcare sector in India. American Journal of Business 38 (3), 152–171.
Kamble, S.S., Gunasekaran, A., Kumar, V., Belhadi, A., Foropon, C., 2021. A machine learning based approach for predicting blockchain adoption in supply chain. Technol. Forecast. Soc. Chang. 163 (2), 1–18.
Kant, N., 2021. Blockchain: a strategic resource to attain and sustain competitive advantage. International Journal of Innovation Science 13 (4), 520–538.
Kashem, M.A., Shamsuddoha, M., Nasir, T., Chowdhury, A.A., 2023. The role of artificial intelligence and blockchain technologies in sustainable tourism in the Middle East. Worldwide Hospitality and Tourism Themes 15 (2), 178–191.
Katebi, A., Homami, P., Najmeddin, M., 2022. Acceptance model of precast concrete components in building construction based on technology acceptance model (TAM) and technology, organization, and environment (TOE) framework. Journal of Building Engineering 45 (1), 1–17.
Kaur, P., Dhir, A., Ray, A., Bala, P.K., Khalil, A., 2021. Innovation resistance theory perspective on the use of food delivery applications. J. Enterp. Inf. Manag. 34 (6), 1746–1768.
Kautish, P., Siddiqui, M., Siddiqui, A., Sharma, V., Alshibani, S.M., 2023. Technology-enabled cure and care: an application of innovation resistance theory to telemedicine apps in an emerging market context. Technol. Forecast. Soc. Chang. 192 (7), 1–16.
Khanna, P., Haldar, A., 2022. Will adoption of blockchain technology be challenging: evidence from Indian banking industry. Qualitative Research in Financial Markets 15 (2), 361–384.
Khaw, K. W., Alnoor, A., Al-Abrrow, H., Chew, X., Sadaa, A. M., Abbas, S., & Khattak, Z. Z. (2022). Modelling and evaluating trust in mobile commerce: a hybrid three stage fuzzy Delphi, structural equation modeling, and neural network approach. International Journal of Human–Computer Interaction, 38(16), 1529–1545.
Khaw, K.W., Sadaa, A.M., Alnoor, A., Zaidan, A.S., Ganesan, Y., Chew, X., 2023. Spurring sustainability commitment strategy of family-owned SMEs: a multi-analytical SEM & ANFIS perspective. J. High Technol. Managem. Res. 34 (1), 1–16.
Kim, H.-W., Kankanhalli, A., 2009. Investigating user resistance to information systems implementation: a status quo bias perspective. MIS Q. 33 (3), 567–582.
Kleijnen, M., Lee, N., Wetzels, M., 2009. An exploration of consumer resistance to innovation and its antecedents. J. Econ. Psychol. 30 (3), 344–357.
Knott, A.M., Vieregger, C., 2020. Reconciling the firm size and innovation puzzle. Organ. Sci. 31 (2), 477–488.
Komulainen, R., Nätti, S., 2023. Barriers to blockchain adoption: empirical observations from securities services value network. J. Bus. Res. 159 (4), 1–12.
Kouhizadeh, M., Saberi, S., Sarkis, J., 2021. Blockchain technology and the sustainable supply chain: theoretically exploring adoption barriers. Int. J. Prod. Econ. 231 (1), 1–21.
Krichen, M., Ammi, M., Mihoub, A., Almutiq, M., 2022. Blockchain for modern applications: a survey. Sensors 22 (14), 1–27.
Kuan, K.K.Y., Chau, P.Y.K., 2001. A perception-based model for EDI adoption in small businesses using a technology–organization–environment framework. Inf. Manag. 38 (8), 507–521.
Kumar, A., Shankar, A., Tiwari, A.K., Hong, H.J., 2023. Understanding Dark Side of Online Community Engagement: An Innovation Resistance Theory Perspective. Article in press, Information Systems and e-Business Management, pp. 1–27.
Kurani, A., Doshi, P., Vakharia, A., Shah, M., 2023. A comprehensive comparative study of artificial neural network (ANN) and support vector machines (SVM) on stock forecasting. Ann. Data Sci. 10 (1), 183–208.
Leong, L.Y., Hew, T.S., Ooi, K.B., Chong, A.Y.L., 2020. Predicting the antecedents of trust in social commerce–a hybrid structural equation modeling with neural network approach. J. Bus. Res. 110 (1), 24–40.
Lin, H.F., Lin, S.M., 2008. Determinants of e-business diffusion: a test of the technology diffusion perspective. Technovation 28 (3), 135–145.
Lu, L., Liang, C., Gu, D., Ma, Y., Xie, Y., Zhao, S., 2021. What advantages of blockchain affect its adoption in the elderly care industry? A study based on the technology–organisation–environment framework. Technol. Soc. 67 (11), 1–10.
Mahmood, A., Arshad Ali, A., Nazam, M., Nazim, M., 2021. Developing an interplay among the psychological barriers for the adoption of industry 4.0 phenomenon. PLoS One 16 (8), 1–26.
Malik, S., Chadhar, M., Vatanasakdakul, S., Chetty, M., 2021. Factors affecting the organizational adoption of blockchain technology: extending the technology–organization–environment (TOE) framework in the Australian context. Sustainability 13 (16), 1–31.
Mention, A.L., 2011. Co-operation and co-opetition as open innovation practices in the service sector: which influence on innovation novelty? Technovation 31 (1), 44–53.
Mioduser, D., Betzer, N., 2008. The contribution of project-based-learning to high-achievers' acquisition of technological knowledge and skills. Int. J. Technol. Des. Educ. 18 (1), 59–77.
Montecchi, M., Plangger, K., Etter, M., 2019. It's real, trust me! Establishing supply chain provenance using blockchain. Bus. Horiz. 62 (3), 283–293.

Morgan, K., 1970. Sample size determination using Krejcie and Morgan table. Kenya Projects Organization (KENPRO) 38 (1), 607–610.
Munim, Z.H., Balasubramaniyan, S., Kouhizadeh, M., Hossain, N.U.I., 2022. Assessing blockchain technology adoption in the Norwegian oil and gas industry using Bayesian best worst method. J. Ind. Inf. Integr. 28 (7), 1–14.
Muzammal, M., Qu, Q., Nasrulin, B., 2019. Renovating blockchain with distributed databases: an open source system. Futur. Gener. Comput. Syst. 90, 105–117.
Nakamoto, S., 2008. Bitcoin: A Peer-to-Peer Electronic Cash System. Satoshi Nakamoto. www.bitcoin.org.
Ngah, A.H., Thurasamy, R., Mohd Salleh, N.H., Jeevan, J., Md Hanafiah, R., Eneizan, B., 2022. Halal transportation adoption among food manufacturers in Malaysia: the moderated model of technology, organization and environment (TOE) framework. Journal of Islamic Marketing 13 (12), 2563–2581.
Nkhoma, M.Z., Dang, D.P., De Souza-Daw, A., 2013. Contributing factors of cloud computing adoption: a technology-organisation-environment framework approach. Proceedings of the European Conference on Information Management & Evaluation 2 (1), 180–188.
Oliveira, T., Thomas, M., Baptista, G., Campos, F., 2016. Mobile payment: understanding the determinants of customer adoption and intention to recommend the technology. Comput. Hum. Behav. 61 (2016), 404–414.
Parasuraman, A., Grewal, D., 2000. The impact of technology on the quality-value-loyalty chain: a research agenda. J. Acad. Mark. Sci. 28 (1), 168–174.
Premkumar, G., Roberts, M., 1999. Adoption of new information technologies in rural small businesses. Omega 27 (4), 467–484.
Radanović, I., Likić, R., 2018. Opportunities for use of blockchain technology in medicine. Appl. Health Econ. Health Policy 16 (1), 583–590.
Ram, S., 1987. A model of innovation resistance. Adv. Cons. Res. 14 (1), 208–212.
Ram, S., Sheth, J.N., 1989. Consumer resistance to innovations: the marketing problem and its solutions. J. Consum. Mark. 6 (2), 5–14.
Rogers, E.M., 2003. Diffusion of Innovations. A Division of Simon & Schuster, New York.
Rogers, E.M., Singhal, A., Quinlan, M.M., 2014. Diffusion of innovations. In: An Integrated Approach to Communication Theory and Research. Routledge, pp. 432–448.
Sadaa, A.M., Ganesan, Y., Khaw, K.W., Alnoor, A., Abbas, S., Chew, X., Bayram, G.E., 2023. Based on the perception of ethics in social commerce platforms: adopting SEM and MCDM approaches for benchmarking customers in rural communities. Curr. Psychol. 42 (35), 31151–31185.
Sadiq, M., Adil, M., Paul, J., 2021. An innovation resistance theory perspective on purchase of eco-friendly cosmetics. J. Retail. Consum. Serv. 59 (3), 1–12.
Samuelson, W., Zeckhauser, R., 1988. Status quo bias in decision making. J. Risk Uncertain. 1 (1), 7–59.
Sayogo, D.S., Zhang, J., Luna-Reyes, L., Jarman, H., Tayi, G., Andersen, D.L., Pardo, T.A., Andersen, D.F., 2015. Challenges and requirements for developing data architecture supporting integration of sustainable supply chains. Inf. Technol. Manag. 16 (1), 5–18.
Senyo, P.K., Effah, J., Addae, E., 2016. Preliminary insight into cloud computing adoption in a developing country. J. Enterp. Inf. Manag. 29 (4), 505–524.
Shahzad, K., Zhang, Q., Ashfaq, M., Zafar, A.U., Ahmad, B., 2024. Pre-to post-adoption of blockchain technology in supply chain management: influencing factors and the role of firm size. Technol. Forecast. Soc. Chang. 198 (1), 1–14.
Shen, B., Dong, C., Minner, S., 2022. Combating copycats in the supply chain with permissioned blockchain technology. Prod. Oper. Manag. 31 (1), 138–154.
Shishan, F., Hmoud, H., Zaidan, H., & Qasem, Z. (2022). The innovation resistance theory: the case of cryptocurrencies. In artificial intelligence for sustainable finance and sustainable technology: proceedings of ICGER 2021 1 (pp. 459-465). Springer international publishing.
Stratopoulos, T.C., Wang, V.X., Ye, H., 2022. Use of corporate disclosures to identify the stage of blockchain adoption. Account. Horiz. 36 (1), 197–220.
Talke, K., Heidenreich, S., 2014. How to overcome pro-change bias: incorporating passive and active innovation resistance in innovation decision models. J. Prod. Innov. Manag. 31 (5), 894–907.
Talwar, M., Corazza, L., Bodhi, R. and Malibari, A. (2023), "why do consumers resist digital innovations? An innovation resistance theory perspective", international journal of emerging markets, Vol. ahead-of-print no. ahead-of-print, 1-16.
Tan, T.M., Salo, J., 2023. Ethical marketing in the blockchain-based sharing economy: theoretical integration and guiding insights. J. Bus. Ethics 183 (4), 1113–1140.
Tao, H., Hu, Y., Li, H., Fan, D., Chen, H., 2022. The credibility measurement model of food safety on-chain data based on blockchain. J. Inter. Technol. 23 (4), 719–725.
Tapscott, D., Tapscott, A., 2017. How blockchain will change organizations. MIT Sloan Manag. Rev. 58 (2), 10–21.
Tariq, B., Najam, H., Han, H., Sadaa, A.M., Abbasi, A.A., Christopher, N., Abbasi, G.A., 2021. Examining mobile financial services in Pakistan: rural and urban perspective with gender as a moderator. Studies in Systems, Decision and Control 335 (1), 225–245.
Thompson, B.S., Rust, S., 2023. Blocking blockchain: examining the social, cultural, and institutional factors causing innovation resistance to digital technology in seafood supply chains. Technol. Soc. 73 (3), 1–11.
Tornatzky, L.G., Fleischer, M., 1990. The Processes of Technological Innovation. Lexington. Lexington Books, MA.
Tseng, F.M., Liang, C.W., Nguyen, N.B., 2023. Blockchain technology adoption and business performance in large enterprises: a comparison of the United States and China. Technol. Soc. 73 (5), 1–9.
Walsh, C., O'Reilly, P., Gleasure, R., McAvoy, J., O'Leary, K., 2021. Understanding manager resistance to blockchain systems. Eur. Manag. J. 39 (3), 353–365.
Wang, Y.Y., Tao, F., Wang, J., 2022. Information disclosure and blockchain technology adoption strategy for competing platforms. Inf. Manag. 59 (7), 1–11.
Wong, L.W., Tan, G.W.H., Lee, V.H., Ooi, K.B., Sohal, A., 2023. Psychological and system-related barriers to adopting Blockchain for operations management: an artificial neural network approach. IEEE Trans. Eng. Manag. 70 (1), 67–81.
Wong, L.-W., Leong, L.-Y., Hew, J.-J., Tan, G.W.-H., Ooi, K.-B., 2020. Time to seize the digital evolution: adoption of blockchain in operations and supply chain management among Malaysian SMEs. Int. J. Inf. Manag. 52 (6), 1–19.
XinYing, C., Tiberius, V., Alnoor, A., Camilleri, M., & Khaw, K. W. (2024). The dark side of metaverse: a multi-perspective of deviant behaviors from PLS-SEM and fsQCA findings. International Journal of Human–Computer Interaction, (Article in press, 1-21).
Xu, B., Li, H., Zhang, X., Alejandro, T.B., 2023. Equilibrium blockchain adoption strategies for duopolistic competitive platforms with network effects. J. Bus. Res. 164 (9), 1–12.
Yli-Huumo, J., Ko, D., Choi, S., Park, S., Smolander, K., 2016. Where is current research on blockchain technology?—a systematic review. PLoS One 11 (10), 1–27.
Yuen, K.F., Cai, L., Qi, G., Wang, X., 2021. Factors influencing autonomous vehicle adoption: an application of the technology acceptance model and innovation diffusion theory. Technology Analysis and Strategic Management 33 (5), 505–519.
Zaidan, A.A., Alnoor, A., Albahri, O.S., Mohammed, R.T., Alamoodi, A.H., Albahri, A.S., Malik, R.Q., 2023. Review of artificial neural networks-contribution methods integrated with structural equation modeling and multi-criteria decision analysis for selection customization. Eng. Appl. Artif. Intell. 124 (9), 1–27.
Zhou, Y., Soh, Y.S., Loh, H.S., Yuen, K.F., 2020. The key challenges and critical success factors of blockchain implementation: policy implications for Singapore's maritime industry. Mar. Policy 122 (12), 1–10.
Zhu, K., Kraemer, K.L., Zhu, K., Kraemer, K.L., 2005. Post-adoption variations in usage and value of E-business by organizations : cross-country evidence from the retail industry post-adoption variations in usage and value of E-business by organizations : cross-country evidence from the retail industry. Inf. Syst. Res. 16 (1), 61–84.

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